KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Yearly Exam Help Sheet

Year 8 Mathematics · Equations, Investigating Data and Mixed Topics

Use this sheet while you work through the practice exams. Every skill on the **Skills Checklist** is here, and the question numbers (Q1–Q22) match the practice exams and the yearly exam. **Exam tips:** about one minute per mark · show your working · check the scale, key and units on every graph · underline words like *at least*, *more than*, *altogether*.

Section A – Equations

Inverse operations

Skill 1 · Q1

To get the pronumeral on its own, **do the opposite** to both sides.

If you see	+	−	×	÷
Do	−	+	÷	×

```
a + 11 = 26        k − 4 = 13
a = 26 − 11        k = 13 + 4
a = 15             k = 17

5w = 45            r ÷ 3 = 12
w = 45 ÷ 5         r = 12 × 3
w = 9              r = 36
```

Two-step equations

Skill 2 · Q2

1. Undo the **+ or −** first.
2. Then undo the **× or ÷**.

```
2x + 9 = 25          x ÷ 4 − 2 = 3
  2x = 25 − 9          x ÷ 4 = 3 + 2
  2x = 16              x ÷ 4 = 5
   x = 8                 x = 5 × 4 = 20
```

Watch out: write each step on a new line – each step can earn a mark.

Negative solutions

Skill 3 · Q3

Solve exactly the same way. If you take a bigger number from a smaller one, the answer is negative.

```
4k + 23 = 3
  4k = 3 − 23 = −20
   k = −20 ÷ 4 = −5
```

Watch out: a negative ÷ a positive gives a negative.

Checking a solution (substitution)

Skill 4 · Q4

1. Replace the pronumeral with the value.
2. Work out the left-hand side.
3. Compare with the right-hand side and write a conclusion.

```
Is a = 5 the solution to 7a − 3 = 32?
  LHS = 7 × 5 − 3 = 35 − 3 = 32
  RHS = 32
  LHS = RHS, so YES, a = 5 is the solution.
```

"Think of a number" problems

Skills 5, 7 · Q5

Write the operations **in the order they happen**, then put "= answer".

Words	Write
multiply by 7	7n
divide by 3	n ÷ 3
add 5 / increase by 5	+ 5
subtract 2 / take away 2	− 2
the answer is 40	= 40

```
"Multiply by 7 then subtract 2. The answer is 40."
  7n − 2 = 40  →  7n = 42  →  n = 6
```

Real-life equations

Skills 6, 7 · Q6

Total = (price each × how many) + fixed amount

```
A gym charges $30 to join plus $12 per week.
Oli paid $114. Let w = number of weeks.
  12w + 30 = 114
  12w = 84
    w = 7   → Oli was a member for 7 weeks.
```

Watch out: finish with a sentence and units ($, hours, students).

Section B – Investigating Data

Reading any graph

Skills 8–11

- Read the **title**, axis labels and units first.
- Check the **scale** – how much is each grid line worth?
- Use a ruler to line up the top of a column or a point with the axis.

Column graphs

Skill 8 · Q7

- **Most popular** = tallest column.
- **Altogether / total** = add the heights of **every** column.

```
Heights 6, 10, 3, 8  →  total = 6 + 10 + 3 + 8 = 27
```

Line graphs

Skill 9 · Q8

Change = later value − earlier value. A rise is an increase, a fall is a decrease.

```
9 am: 7°C,  3 pm: 19°C  →  rise = 19 − 7 = 12°C
10 pm: 2°C,  5 am: −3°C →  fall = 2 − (−3) = 5°C
```

Watch out: subtracting a negative is the same as adding.

Sector (pie) graphs

Skills 10, 11 · Q9

Amount = **percentage × total**. To change a % to a decimal, divide by 100.

```
45% of 80 = 0.45 × 80 = 36
```

How many more: subtract the percentages first, then find that % of the total.

```
Soccer 45%, Tennis 15%, total 80
  45% − 15% = 30%   →  0.3 × 80 = 24 more
```

Stem-and-leaf plots

Skills 12, 13 · Q10

Use the **key**: if 3 | 6 means 36, the stem is the tens and each leaf is the units. Each leaf is one value, so **count the leaves** to find how many values.

Stem	Leaf
5	1 4 4
6	0 2 7 8

Values: 51, 54, 54, 60, 62, 67, 68 (7 values → median is the 4th = 60).

Dot plots

Skill 14 · Q11

- Each dot is **one** person or result.
- **Mode** = the value with the tallest stack of dots.
- Total = count every dot.

"At least" and "more than"

Skill 15 · Q10–Q12

Words	Means	Example with 4
at least 4	4 or more (≥)	4, 5, 6, …
more than 4	bigger than 4 (>)	5, 6, 7, …
at most 4	4 or less (≤)	…, 3, 4
fewer than 4	smaller than 4 (<)	…, 2, 3

Watch out: "more than 40" does NOT include 40 itself.

Frequency tables

Skill 16 · Q12

Score	1	2	3	4
Frequency	5	12	7	2

- **Total surveyed** = add the frequencies: 5 + 12 + 7 + 2 = 26
- **Mode** = the score with the highest frequency = **2** (not 12!)
- **At least 3** = 7 + 2 = 9

Mean, median, mode and range

Skills 17–19 · Q13

Mean	add all the values ÷ how many values
Median	put in order, find the middle value
Mode	the value that appears most often
Range	highest − lowest

```
Data: 7, 3, 10, 4, 10, 6   (6 values)
Mean   = 40 ÷ 6 = 6.67 (2 d.p.)
Order  = 3, 4, 6, 7, 10, 10
Median = (6 + 7) ÷ 2 = 6.5   ← even number of values
Mode   = 10        Range = 10 − 3 = 7
```

Watch out: for an even number of values the median is halfway between the two middle values.

Outliers

Skill 20 · Q14

An **outlier** is a value much bigger or much smaller than the rest.

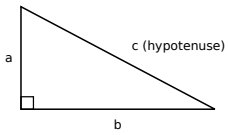
```
Data: 21, 19, 22, 20, 78
78 is the outlier.
Mean with it    = 160 ÷ 5 = 32
Mean without it = 82 ÷ 4  = 20.5
```

A large outlier pulls the mean **up**; a small outlier pulls it **down**. The median hardly changes.

Section C – Mixed Topics

Pythagoras' theorem

Skills 21, 22 · Q15



$$c^2 = a^2 + b^2$$

The **hypotenuse** c is the longest side, opposite the right angle.

```
Find c when a = 5 and b = 12:
  c² = 5² + 12² = 25 + 144 = 169
  c  = √169 = 13
```

Is it right-angled? Square the two short sides and add. If it equals the longest side squared → YES.

```
9, 14, 17:  9² + 14² = 81 + 196 = 277
            17² = 289   277 ≠ 289  → NO
```

Decimals

Skill 23 · Q16

Multiplying: ignore the decimal points, multiply, then put back the total number of decimal places.

```
0.3 × 0.07  →  3 × 7 = 21
1 d.p. + 2 d.p. = 3 d.p.  →  0.021
```

Dividing by a decimal: move the point in **both** numbers the same number of places until you divide by a whole number.

```
6.4 ÷ 0.08  =  640 ÷ 8  =  80
```

Watch out: check the answer makes sense – dividing by a number less than 1 makes the answer bigger.

Powers and square roots

Skill 24 · Q17

3^4 means 3 × 3 × 3 × 3 = 81 (not 3 × 4). √ means "what number times itself gives this?" e.g. √169 = 13

n	1	2	3	4	5	6	7	8	9	10	11	12
n^2	1	4	9	16	25	36	49	64	81	100	121	144
n^3	1	8	27	64	125	216	343	512	729	1000		

Also: $13^2 = 169$, $14^2 = 196$, $15^2 = 225$, $20^2 = 400$

Ratios and rates

Skills 33, 34 · Q22

Simplify a ratio: divide both parts by the highest common factor (HCF).

```
16 : 40  → HCF is 8  → 2 : 5
```

Average speed = distance ÷ time

```
260 km in 4 hours → 260 ÷ 4 = 65 km/h
```

Watch out: if the time is in minutes, change it to hours first: 30 min = 0.5 h.

Surds

Skills 25, 26 · Q18

A **surd** is a square root that is **not** a whole number. √3, √10, √40 are surds. √36 = 6 is not.

Simplify: find the **largest square factor** (4, 9, 16, 25, 36, 49, 64, 81, 100).

```
√28  = √4 × √7  = 2√7
√90  = √9 × √10 = 3√10
√200 = √100 × √2 = 10√2
```

Watch out: always use the biggest square factor: √32 = √16 × √2 = 4√2.

Algebra

Skills 27–29 · Q19

Like terms have exactly the same pronumeral part (x and x^2 are NOT like terms).

```
5p + 2q − 3p + q = 2p + 3q
```

Expand: multiply the number outside by **every** term inside.

```
7(k − 2) = 7k − 14
−3(y + 4) = −3y − 12
```

Substitute: swap pronumerals for numbers, then use BODMAS. Powers before ×.

```
4a² − b, a = 3, b = 10
= 4 × 3² − 10 = 4 × 9 − 10 = 26
```

Watch out: $4a^2$ means $4 \times (a^2)$, not $(4a)^2$. Put negatives in brackets: $(-3)^2 = 9$.

Area and volume

Skills 30, 31 · Q20

Trapezium: A = ½ × (a + b) × h

```
a = 4 cm, b = 12 cm, h = 5 cm
A = ½ × (4 + 12) × 5 = ½ × 16 × 5 = 40 cm²
```

Rectangular prism: V = l × w × h

```
V = 7 × 3 × 4 = 84 cm³
```

Watch out: area is in units² (cm²), volume is in units³ (cm³). h is the perpendicular height.

Tables of values

Skill 32 · Q21

Substitute each x-value into the rule. Use brackets for negatives.

```
y = 2x + 5
x = −3: y = 2 × (−3) + 5 = −6 + 5 = −1
x = 0:  y = 2 × 0 + 5 = 5
```

x	−3	−2	−1	0
y	−1	1	3	5

Watch out: the y-values should go up (or down) by the same amount each time – use this to check.